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Unit – 4 : Session – 2 : SLO – 1 | Odd Man Out — ALU Operations

■ I. Odd Man Out

1. Group ADD, SUBTRACT, MULTIPLY, FETCH → FETCH — it is an instruction-cycle step, not an ALU arithmetic operation.
2. Group AND, OR, NOT, LOAD → LOAD — it is a data-transfer operation, not a logical operation.
3. Group SHIFT LEFT, SHIFT RIGHT, ROTATE, COMPARE → COMPARE — it only sets condition flags; the others physically move bits.
4. Group INCREMENT, DECREMENT, DIVIDE, NEGATE → DIVIDE — it is a binary (two-operand) operation; the rest are unary (single-operand) operations.
5. Group XOR, XNOR, MOV, NAND → MOV — it transfers data; the others are Boolean logic operations.
6. Group CARRY FLAG, ZERO FLAG, OVERFLOW FLAG, CLOCK SIGNAL → CLOCK SIGNAL — it synchronizes the CPU; the others are status/condition flags.
7. Group ARITHMETIC, LOGIC, BRANCH, BITWISE → BRANCH — it is a control-flow operation; the others are ALU data-processing categories.
8. Group REGISTER, ADDER, COMPARATOR, MULTIPLEXER → REGISTER — it is a storage (sequential) element; the others are combinational circuits.
9. Group BINARY OPERATION, UNARY OPERATION, DATA TRANSFER, FLOATING POINT → FLOATING POINT — it is a data format/type, not an operation class.
10. Group BOOTH'S ALGORITHM, RESTORING DIVISION, CACHE HIT, WALLACE TREE → CACHE HIT — it is a memory-hierarchy event; the others are arithmetic hardware algorithms.